

McLeod County as Potential Microcosm: Leveraging a Semi-Rural Community to Track SARS-CoV-2 Transmission

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The purpose of this work

- Phylogenetic & phylodynamic analyses enabled real-time tracking of viral spread, variant emergence, and intervention impacts
- Still, rural/semi-rural regions were undersampled, leaving transmission dynamics in these communities poorly understood
- McLeod County, MN is a rare exception with ~25–30% genomic sampling proportion and rich metadata (age, vaccination status, location)

McLeod, MN: Background Information

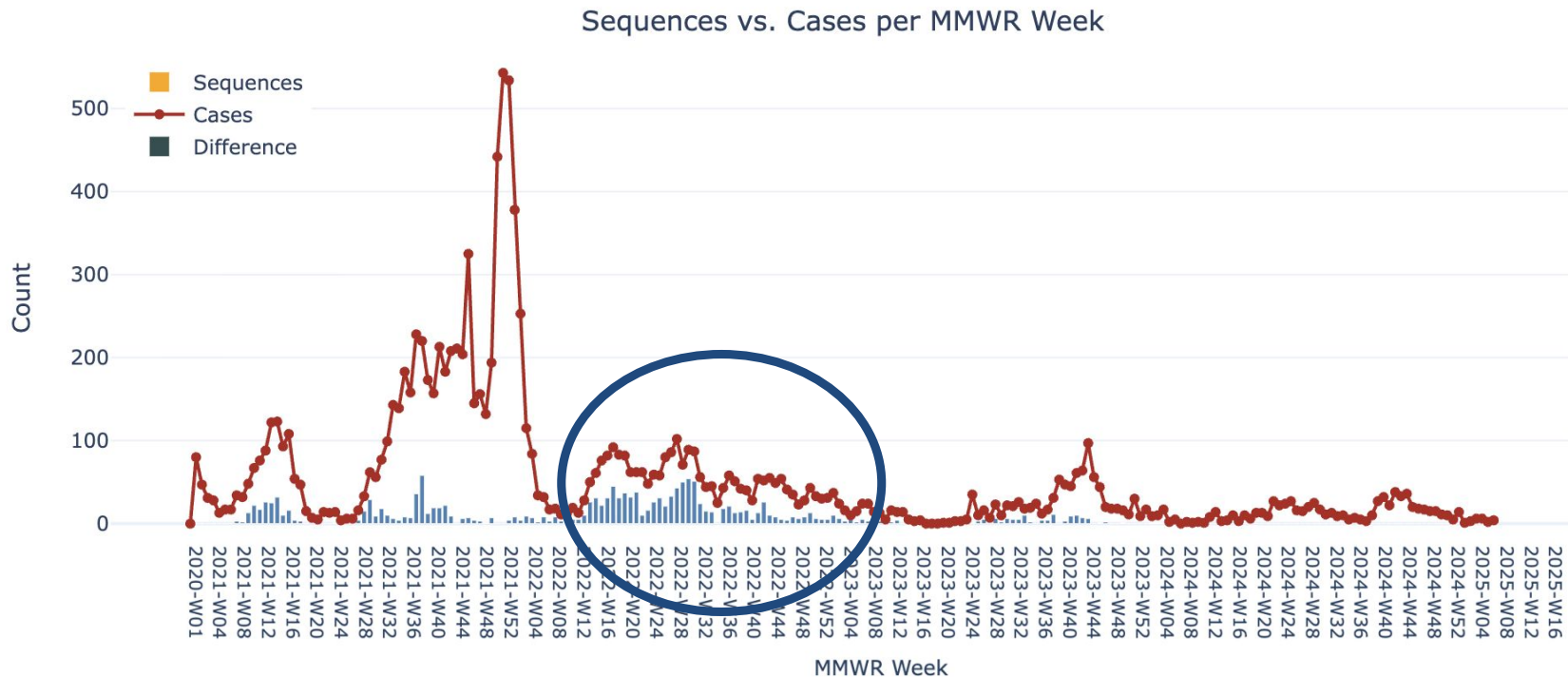


Source: census.gov

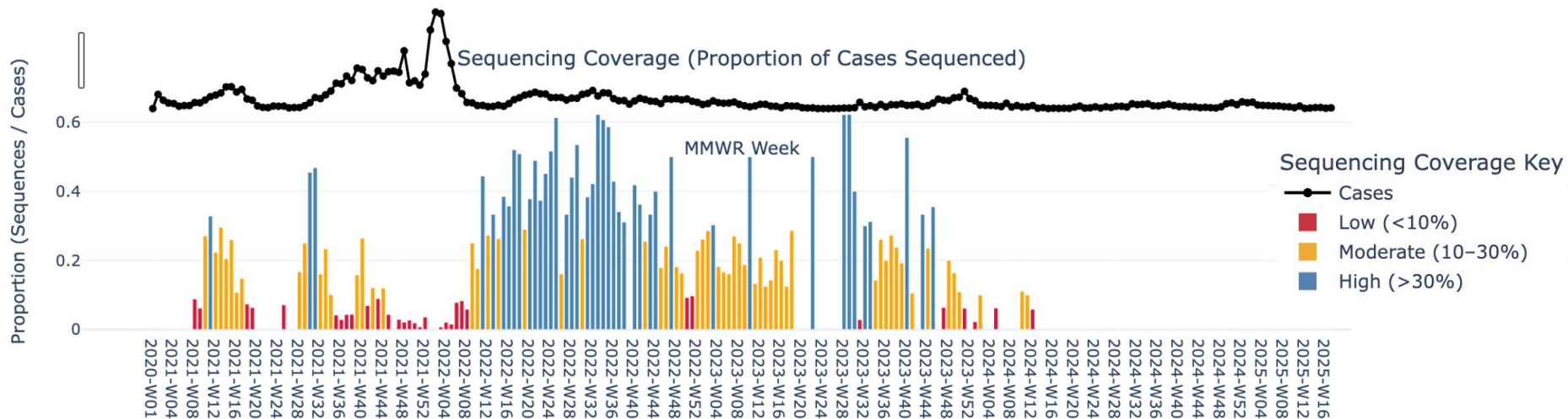
Can a densely sampled, semi-rural community (McLeod County, MN) serve as a reliable microcosm of broader regional and statewide SARS-CoV-2 transmission dynamics?

1. How did sequencing coverage vary across the pandemic timeline?
2. How is transmission in McLeod connected to transmission in the rest of Minnesota?
3. Was McLeod County's sampling proportion (~25-30%) sufficient to capture true local transmission patterns and how many local sequences from McLeod County are needed before inferred introduction counts reliably reflect true transmission dynamics during Omicron?
4. Did the rise and fall of major SARS-CoV-2 variants in McLeod mirror statewide Minnesota frequency trajectories?

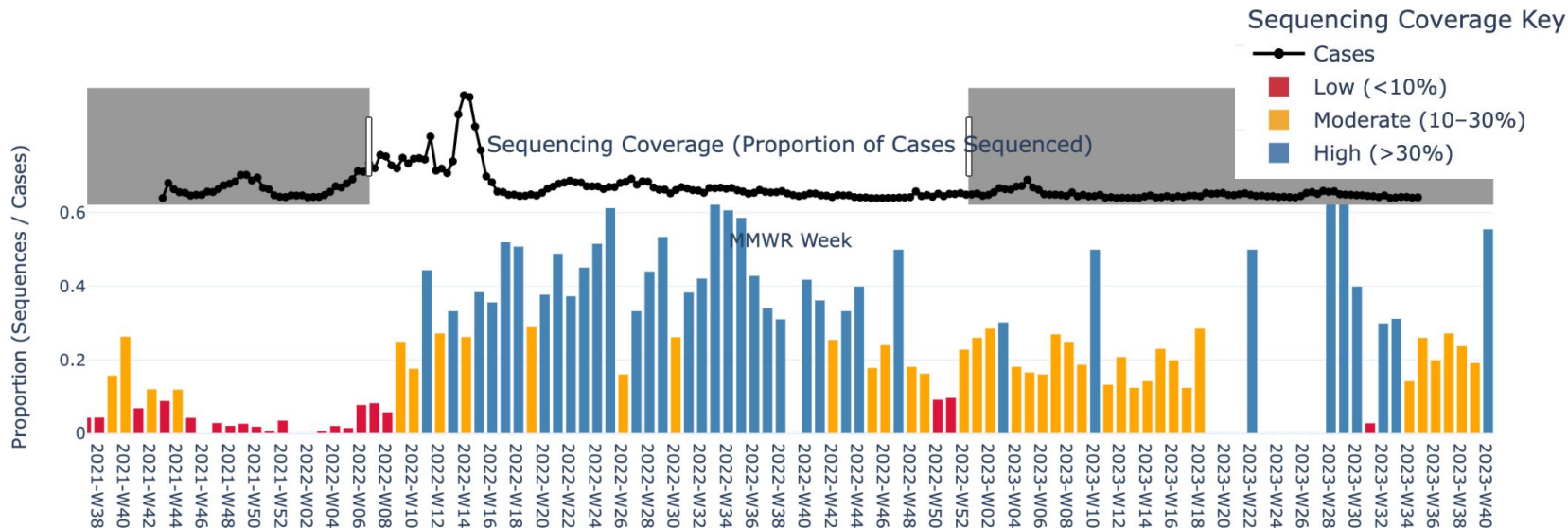
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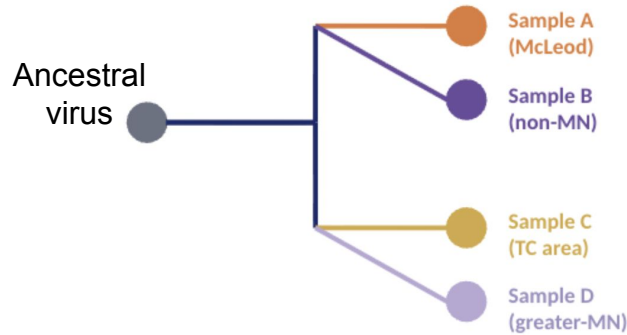


1. How did sequencing coverage vary across the pandemic timeline?



2. How is transmission in McLeod connected to transmission in the rest of Minnesota, and the rest of the country?

In order to answer this question, we need to better understand what a phylogenetic tree is and how it can be used.



Each colored dot = one sequenced virus sample

Earlier → More recent

What the colors mean

Each dot on the tree represents one sequenced virus sample. The color tells us where that sample came from:



McLeod County

Virus samples collected from McLeod County.



Twin Cities Metro (TC County)

Samples collected from Hennepin and Ramsey counties.



Greater Minnesota

Samples from elsewhere in Minnesota, excluding McLeod County, Hennepin County, and Ramsey County.

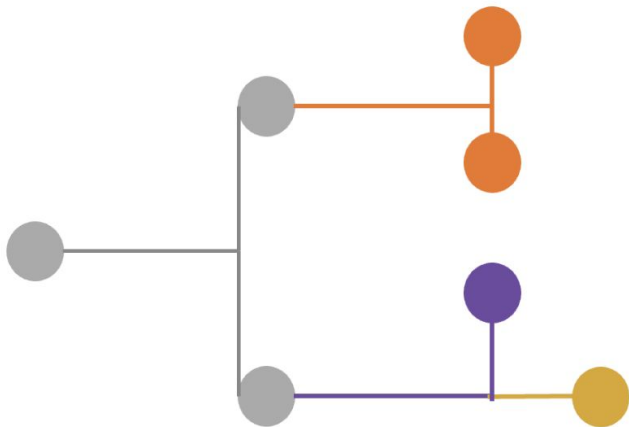


Non-Minnesota (rest of U.S.)

Samples from other states, showing how viruses moved in and out of Minnesota.

When McLeod samples (orange) cluster closely with non-MN samples (purple), it suggests a likely introduction from out of state.

How to read the tree



Node

A dot on the tree. Colored nodes (tips) are real collected virus samples.

Branch

A line connecting two nodes. Longer branches mean more genetic occurred. Shorter branches mean the viruses are very similar genetically.

Cluster

A group of closely related samples that share a recent common ancestor. Samples in the same cluster are likely part of the same chain of transmission.

Host & Inferred Sources of Transmission

What is a Host?

A host is the person (or organism) that a virus lives in and infects.

In this study, each host is a person who tested positive for COVID-19 and whose virus was genetically sequenced.

When we track which host a virus came from, we're asking, "**Who likely gave this virus to whom?**"

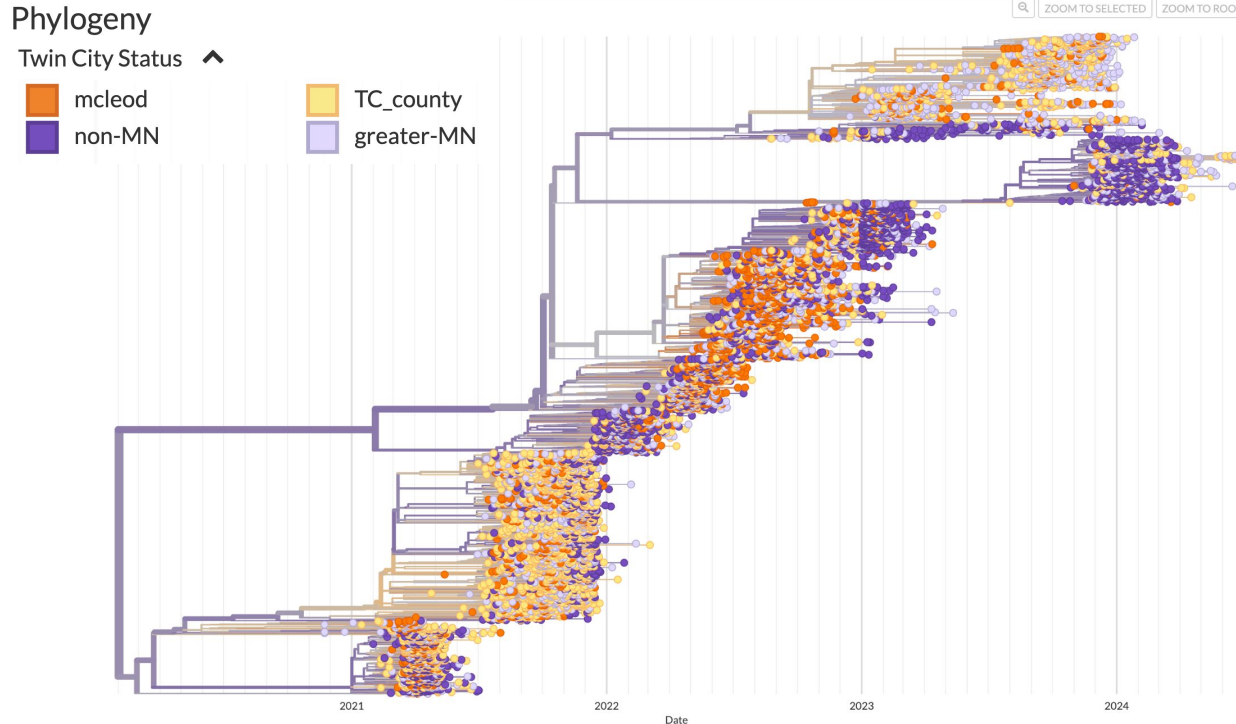
What is Inferred Transmission?

Inferred means we didn't directly observe the virus pass from one person to another, we **estimated it from the genetic data**.

If a McLeod County sample is genetically nearly identical to a sample from Chicago, we infer the virus likely traveled from Chicago to McLeod (or vice versa).

2. How is transmission in McLeod connected to transmission in the rest of Minnesota, and the rest of the country?

Showing 7956 of 7956 genomes sampled between Nov 2020 and Jun 2024.



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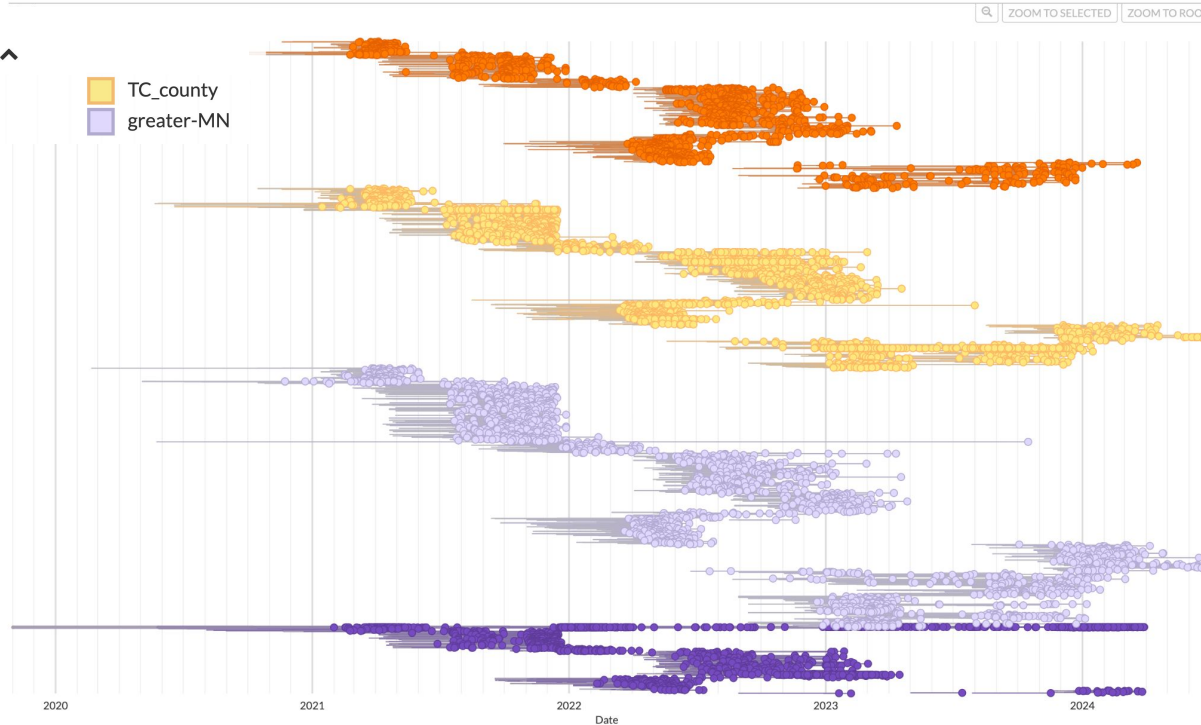
Showing 8019 of 7952 genomes sampled between Nov 2020 and Jun 2024.

Phylogeny

Twin City Status ^

mcLeod
non-MN

TC_county
greater-MN

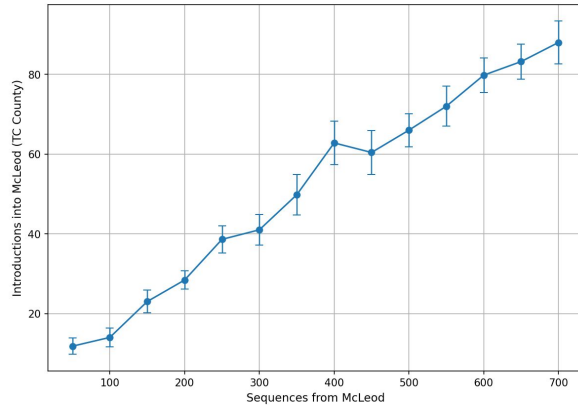


3a. How many local sequences from McLeod County are needed before inferred introduction counts stabilize and reliably reflect true transmission dynamics during Omicron? Rarefaction Methods

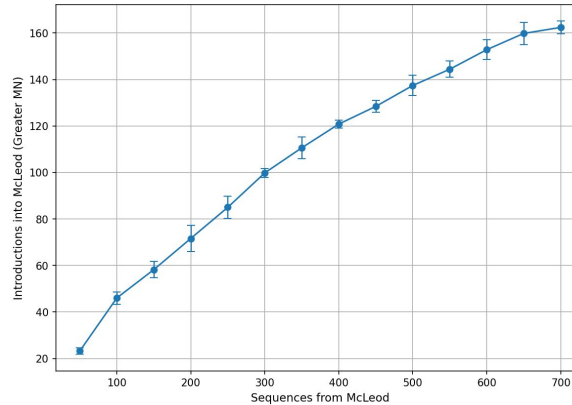
- Selected a fully contextualized wildcard phylogenetic build as the backbone, containing all available contextual sequences from greater-MN (2000 sequences), Twin Cities (2500 sequences), and non-MN (2000 sequences)
- Subsampled McLeod County sequences in incremental steps of 50, for a total of 700 sequences
- Generated 5 independent iterations at each sampling increment to account for variation in sequence selection
- Created a phylogenetic tree for each build (14 sequence increments * 5 iterations)
- Parsed directional viral transitions to count the number of introductions into McLeod County at each sampling level, all done in the same loop
- Plotted introductions into McLeod (y-axis) against number of McLeod sequences included (x-axis)
- Asymptote of the curve identifies the minimum sampling threshold at which introduction counts stabilize

3b. How much data do we need to confidently infer introductions and how do the locations of these vary? Rarefaction Results

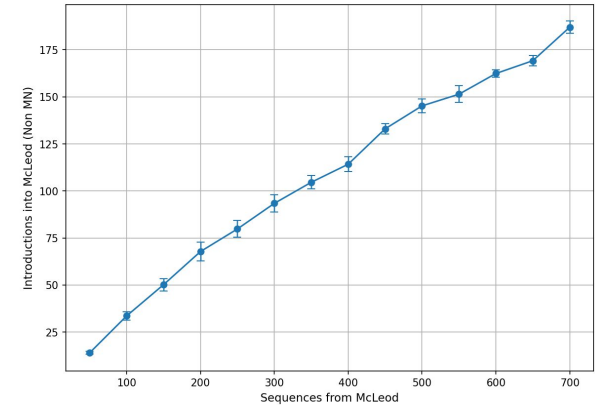
Rarefaction Curve (TC County → McLeod)



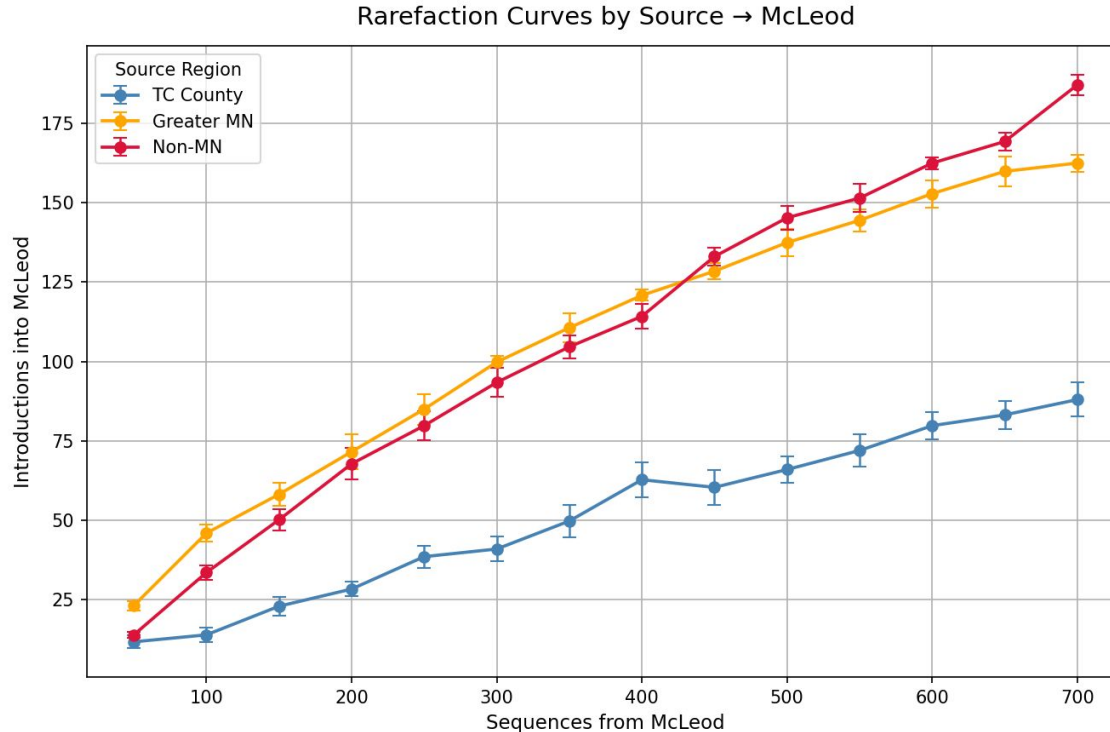
Rarefaction Curve (Greater MN → McLeod)



Rarefaction Curve (Non MN → McLeod)

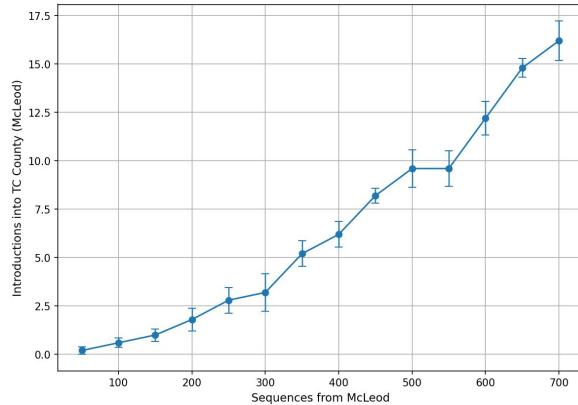


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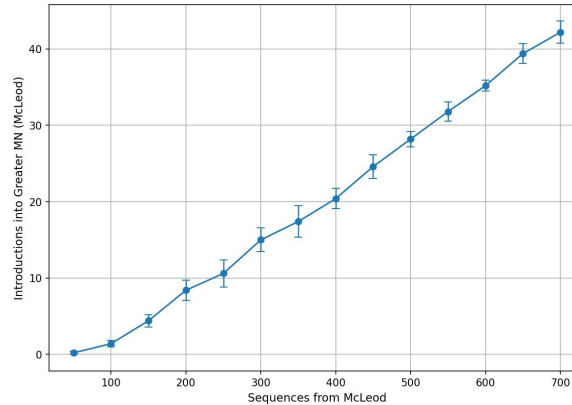


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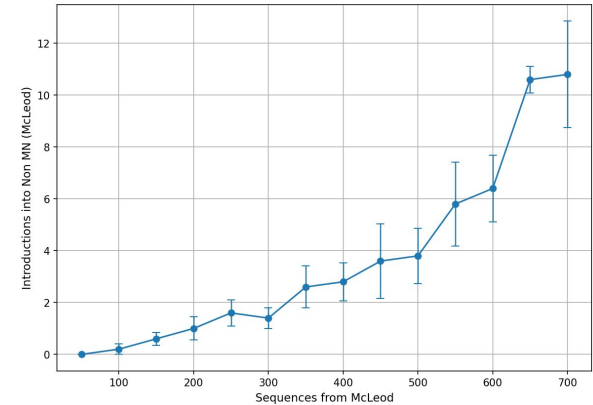
Rarefaction Curve (McLeod → TC County)



Rarefaction Curve (McLeod → Greater MN)

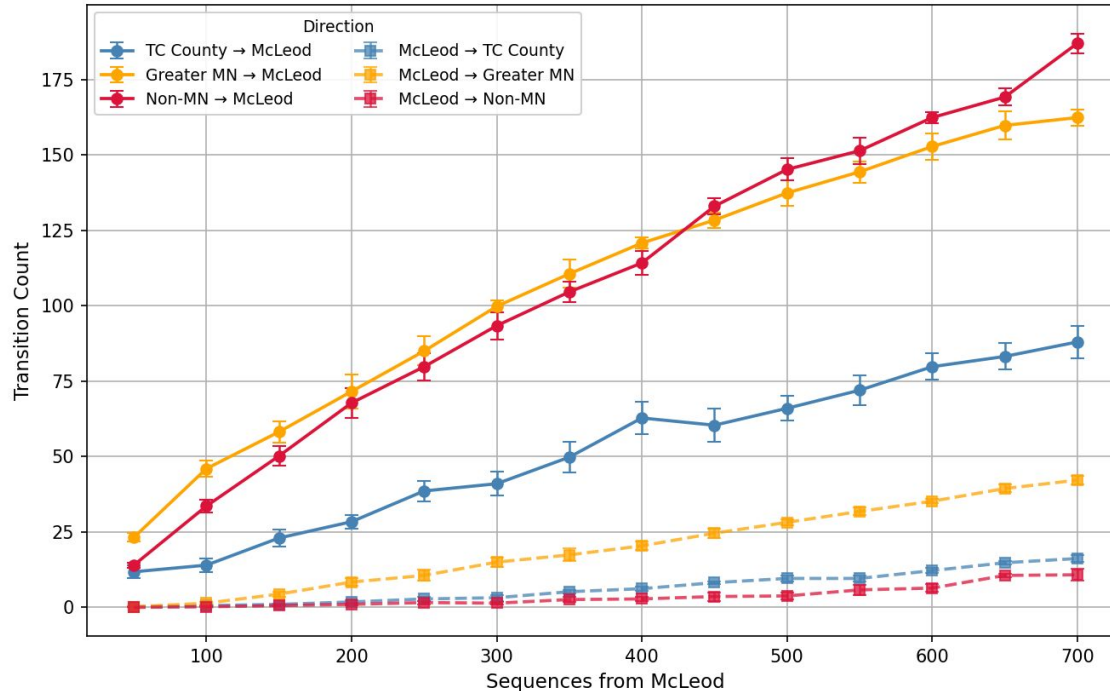


Rarefaction Curve (McLeod → Non MN)

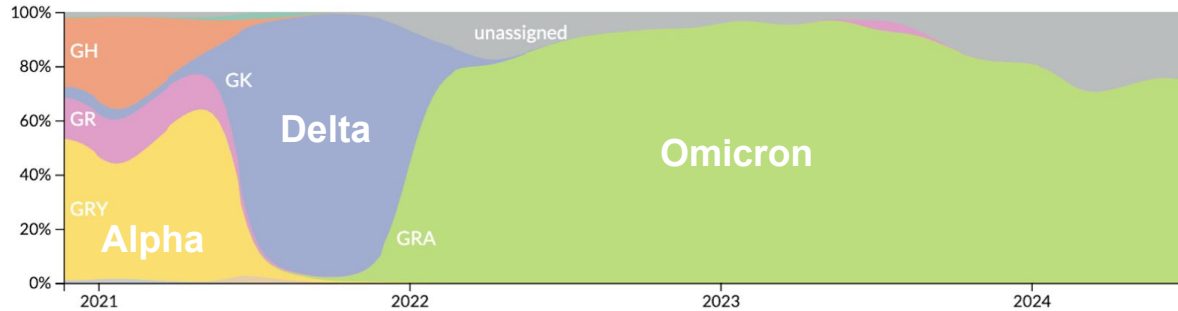


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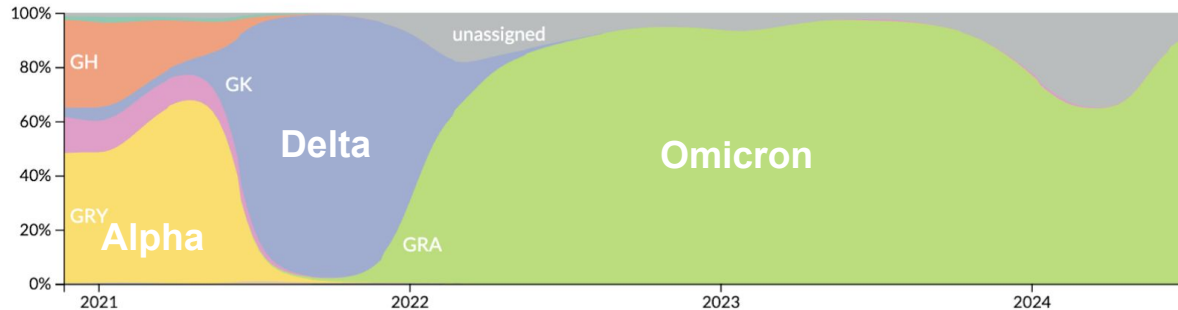
Rarefaction Curves: Introductions Into and Out of McLeod



4. Did the rise and fall of major SARS-CoV-2 variants in McLeod mirror statewide Minnesota frequency trajectories?



McLeod County only colored frequencies by SARS-CoV-2 Strain and normalized to 100% at each time point for 1464 out of a total of 7956 tips (GRA represents Omicron, GK represents Delta, and GRY represents Alpha).

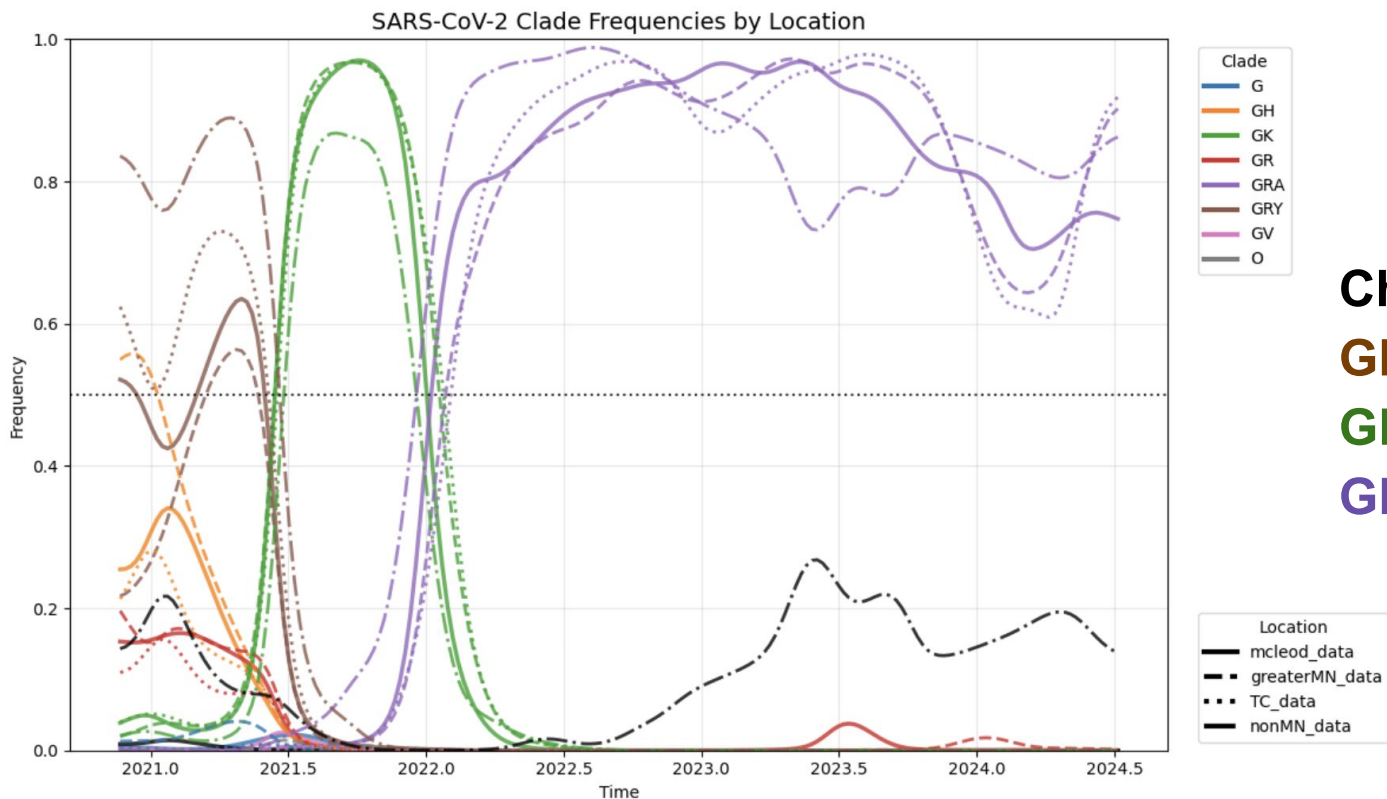


Minnesota only colored frequencies by SARS-CoV-2 Strain and normalized to 100% at each time point for 5958 out of a total of 7956 tips (GRA represents Omicron, GK represents Delta, and GRY represents Alpha).

4. Did the rise and fall of major SARS-CoV-2 variants in McLeod mirror statewide Minnesota frequency trajectories? Variant Frequencies Methods

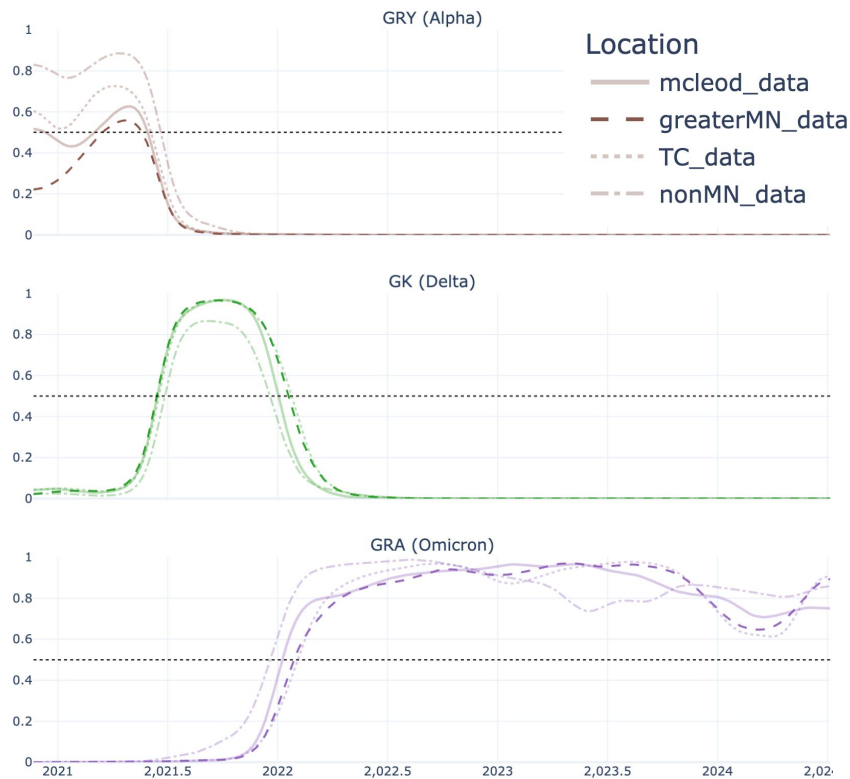
- Generated a fully contextualized Nextstrain build including sequences from McLeod County, greater-MN, Twin Cities, and non-MN states
- Calculated frequencies on the phylogenetic tree to estimate the relative abundance of each SARS-CoV-2 clade at each time point
- Frequencies were computed across four location categories separately: McLeod, greaterMN, TC_data, and nonMN
- Plotted clade frequency (y-axis) against time (x-axis)
- Compared trajectory shapes across location groups to assess whether McLeod's variant dynamics mirrored statewide and regional trends

McLeod County Variant Dynamics Track Statewide Minnesota Trajectories Across the SARS-CoV-2 Pandemic Timeline

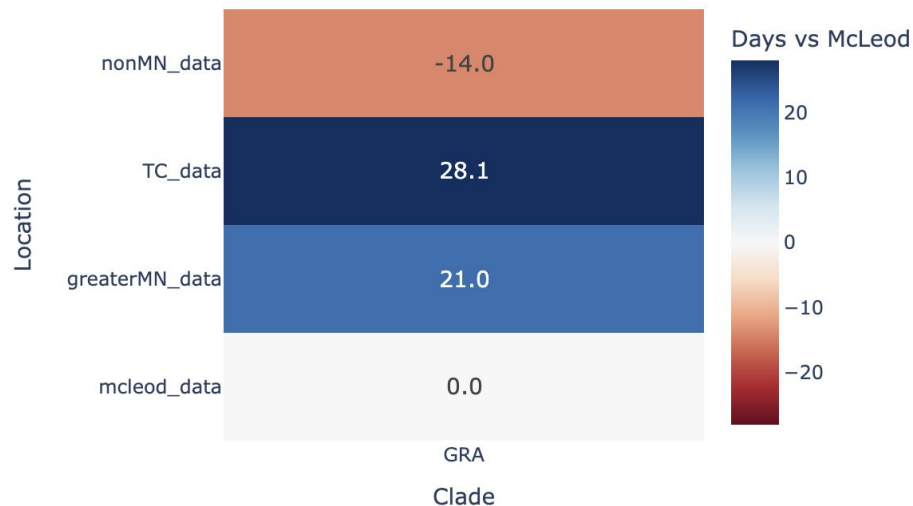


Cheatsheet:
GRY = Alpha
GK = Delta
GRA = Omicron

McLeod County Variant Dynamics Track Statewide Minnesota Trajectories Across Across the SARS-CoV-2 Pandemic Timeline



Omicron (GRA) 50% Crossing Time Lags vs McLeod (Days)



Summary

Can McLeod County serve as a reliable microcosm of broader regional SARS-CoV-2 transmission dynamics?

Q1. Sequencing Coverage

Coverage remained moderate (25–30%) during the Omicron period, with notable dips during high-case-count surges.

Q3. Sampling Sufficiency (Rarefaction)

Introduction counts from all three source regions continued to rise through 700 sequences without clear asymptote, suggesting more sequences improve inference. McLeod exports far fewer introductions than it receives.

Q2. Transmission Connectivity

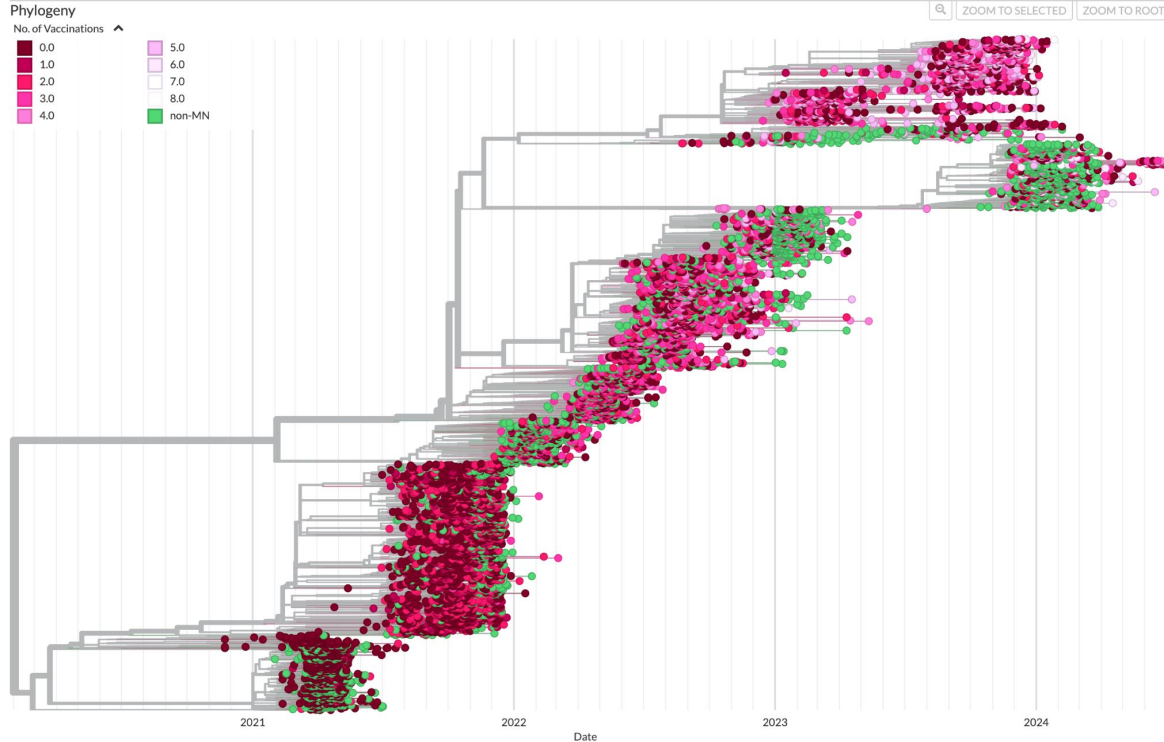
McLeod is deeply embedded in regional transmission networks. Phylogenetic clustering shows frequent mixing with Twin Cities, greater-MN, and non-MN lineages throughout the pandemic.

Q4. Variant Frequency Trajectories

Alpha, Delta, and Omicron rose and fell in McLeod in near-perfect sync with statewide MN trends. McLeod reached the Omicron 50% frequency threshold ~14 days ahead of non-MN, and ~3–4 weeks after TC and greater-MN.

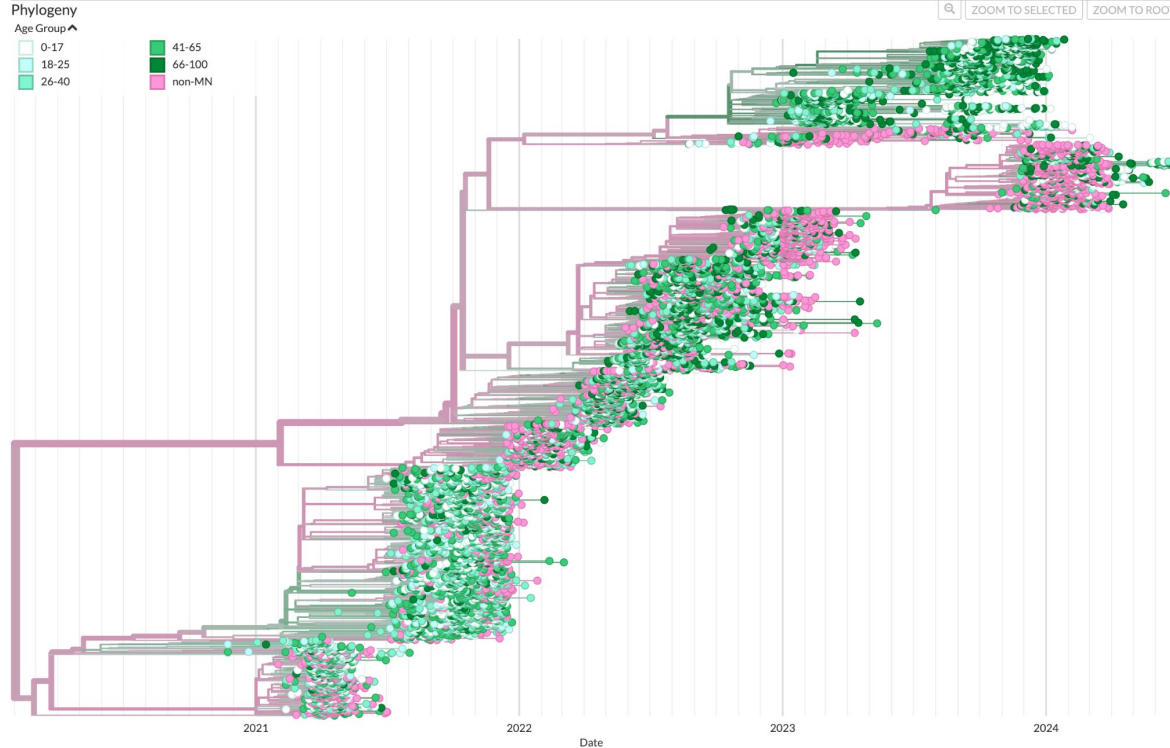
Next Steps and Future Directions

Showing 7956 of 7956 genomes sampled between Nov 2020 and Jun 2024.



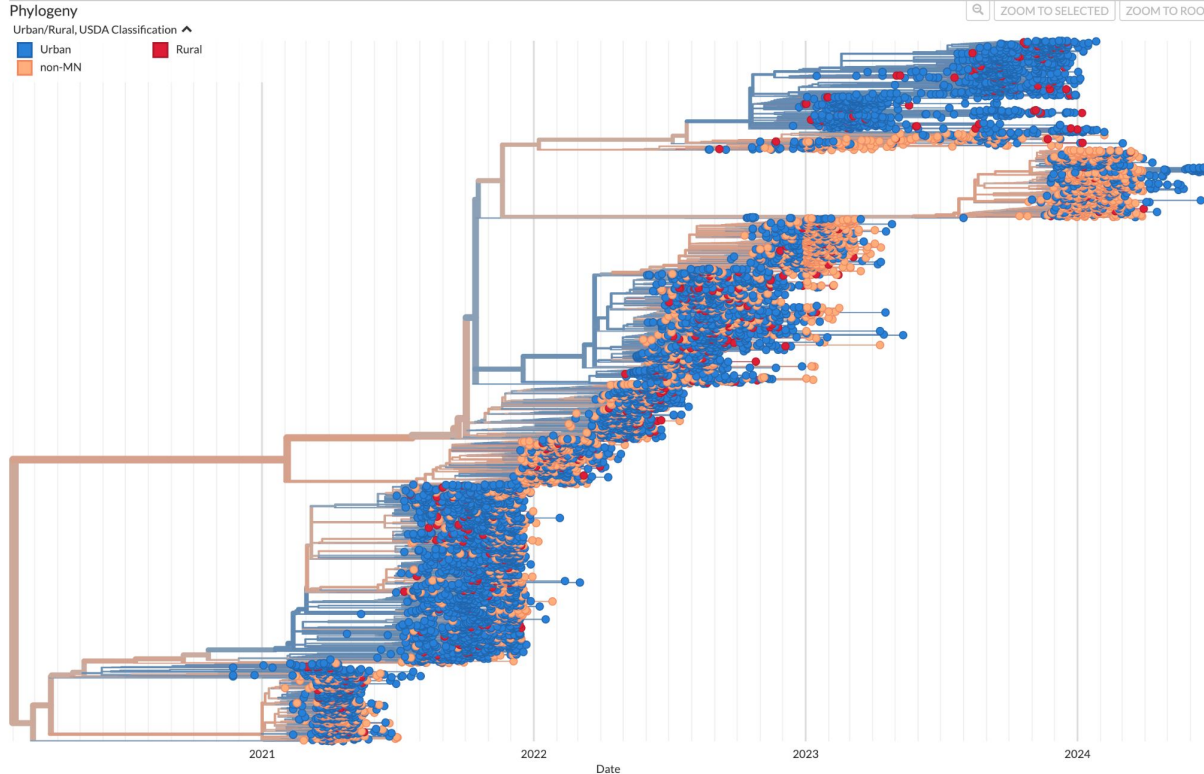
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Next Steps and Future Directions

- Demographic bias and structure in sampling
- Who/where drives introductions (age, vaccination count, and rurality)
- Does this generalize to other counties?
- Sampling thresholds for reliable inference

→ We've shown that dense local sampling in a semi-rural community can capture regional dynamics

Questions and Requests

- How might available demographic data (age, vaccination status, rurality) be leveraged to further characterize transmission dynamics and strengthen our central research question?
- To what extent can findings from McLeod County be generalized to other Minnesota counties with lower genomic sampling proportions?
- Do the sequences in our dataset reflect individuals who resided in McLeod County, or individuals who were tested there?
- Would it be possible to obtain case counts and sequence counts from greater-MN and Twin Cities counties to estimate and compare sampling proportions across regions?

Acknowledgements

Mentorship & Guidance

- Dr. Louise Moncla: project sponsor
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- Minnesota Department of Health (MDH) — data provision

